

NEXUS

The multi-model inference engine for hardware you already own

Free. Local. Portable. Nexus brings real large-language-model performance to a single modest graphics card, an old laptop, or a CPU alone — and links spare computers together when one isn't enough. No subscription, no cloud, nothing leaves your machine.

Multi-model | GPU-accelerated prefill | Persistent KV memory | Multi-computer distributed inference

LOCAL AI · DEMOCRATIZED

◆ WHAT IS NEXUS

One engine, built for the hardware you have

Most local AI tools assume you own a graphics card built for the job. Nexus assumes you don't — and is engineered from the ground up to get real results out of whatever you've actually got.

Run several models at once

Nexus is a **multi-model inference engine** — load more than one model into completely independent slots and run them side by side, each on its own port, each with its own graphics-card placement, sampling settings, and conversation memory. Compare two models head to head, or just keep a small fast model and a larger careful one both warm and ready at the same time.

Prefill acceleration

Reading a long prompt — "prefill" — is normally the slowest, most hardware-hungry part of talking to a large model, and the part that suffers most on modest hardware. Nexus's prefill acceleration streams the model's layers through a graphics card *one at a time* while the prompt is read, then hands off to the CPU (or a GPU-held copy) to generate the reply. The card never needs to hold the whole model — it only ever needs room for a single layer at once.

> The result: GPU-class prompt-reading speed from a card nowhere near big enough to hold the model itself.

◆ HOW IT STAYS FAST

Never pay the same compute cost twice

Two more ideas sit underneath everything Nexus does — one about memory, one about sharing hardware you already own.

Custom KV retention

Most local chat tools re-read the entire conversation from the first token, every single turn. Nexus doesn't. Its KV cache — the model's own working memory of everything it has already read — is retained and reusable. Hand it a huge document, wait once for it to actually be read, and every short follow-up question after that skips the re-read entirely. Save that state to disk, and reloading it drops you straight back into the conversation where you left off — no re-processing, no waiting, just the answer.

Two computers, one model

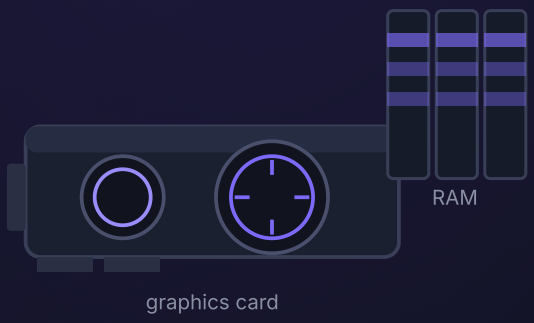
Short on memory for a bigger model, but there's a second computer sitting around doing nothing? Link them together. Nexus splits a single model's layers across two or more machines over the network, turning a pair of ordinary computers into something neither one could run alone.

The benefit isn't only extra capacity. If the main computer has even a single small graphics card — as little as 1 GB — it can still accelerate prefill for the *whole* setup, prompt-reading speed included, even while the second, donor computer has no graphics card at all. The slower machine only has to carry its share of actually generating the reply, where it counts least.

◆ REAL HARDWARE, REAL RESULTS

This isn't a benchmark machine

Everything above isn't theoretical — it's what Nexus's own test rig runs on. A three-year-old desktop graphics card, off-the-shelf memory, and a 27-billion-parameter model that has no business running this well on hardware like this.



THE TEST RIG

MEMORY	25 GB DDR4 @ 2200 MHz
GRAPHICS CARD	3 GB — GeForce GTX 1060
MODEL	Qwen 3.8 27B, Q4
PREFILL SPEED	~50 tok/sec

A 27-billion-parameter model, quantized to 4 bits, reading prompts at roughly 50 tokens per second of prefill — on a 3 GB graphics card that launched in 2016 and 25 GB of ordinary desktop memory. No data-center GPU, no multi-thousand-dollar rig. This is the entire point of Nexus: the hardware most people already own, put to real use.

3 GB VRAM
GRAPHICS CARD MEMORY

27B params
MODEL SIZE (Q4)

~50 tok/s
PREFILL THROUGHPUT

Get started

Nexus is free and runs entirely on your own machine. Pick a model, point Nexus at it, and you're generating — no account, no cloud round-trip, nothing phoned home.

Launcher Guide
First run through picking a model, GPU placement, and starting your first chat.

Quant Builder Guide
Converting and packaging your own models into Nexus's native format.

Engine / API Guide
Running headless, the full HTTP API, and linking the engine directly.

Released under AGPLv3.